

Assignment 3

Full-Stack Quiz Master Application

Aim:

To design and develop a full-stack quiz web application using HTML, CSS, Python, and Flask.

Tasks:

Created Flask project structure

Developed backend using Flask (app.py)

Designed HTML pages:

- Home Page
- Quiz Page
- Result Page

Implemented quiz logic with Python

Processed form data and calculated score

Applied CSS styling for UI

Added restart quiz functionality

Tech Stack :

HTML, CSS, Python, Flask, vs code, chrome

Code:

```
app.py | X # style.css | index.html | quiz.html | result.html
app.py > ...
1 from flask import Flask, render_template, request
2 import random
3
4 app = Flask(__name__)
5
6 # 🔥 LARGE QUESTION BANK (add more if you want)
7 quiz_bank = [
8     {"question": "Capital of India?", "options": ["Delhi", "Mumbai", "Chennai", "Kolkata"], "answer": "Delhi"},
9     {"question": "2 + 2 = ?", "options": ["3", "4", "5", "6"], "answer": "4"},
10    {"question": "HTML stands for?", "options": ["Hyper Text Markup Language", "High Text Machine", "Hyper T"},
11    {"question": "Python is?", "options": ["Language", "Snake", "Game", "OS"], "answer": "Language"},
12    {"question": "CSS used for?", "options": ["Styling", "Logic", "Database", "Server"], "answer": "Styling"},
13    {"question": "JS used for?", "options": ["Frontend", "Backend", "Both", "None"], "answer": "Both"},
14    {"question": "Flask is?", "options": ["Framework", "Language", "Database", "Tool"], "answer": "Framework"},
15    {"question": "1 byte = ?", "options": ["8 bits", "4 bits", "16 bits", "32 bits"], "answer": "8 bits"},
16    {"question": "CPU stands for?", "options": ["Central Processing Unit", "Control Unit", "Computer Unit"], "answer": "Central Processing Unit"},
17    {"question": "Which is DB?", "options": ["MySQL", "HTML", "CSS", "JS"], "answer": "MySQL"}
18 ]
19
20 # Home
21 @app.route('/')
22 def home():
23     return render_template('index.html')
24
25
26 # Quiz Page (random 5 questions every time)
27 @app.route('/quiz')
28 def quiz():
29     global current_quiz
30
31     # pick 5 random questions
32     current_quiz = random.sample(quiz_bank, 5)
```

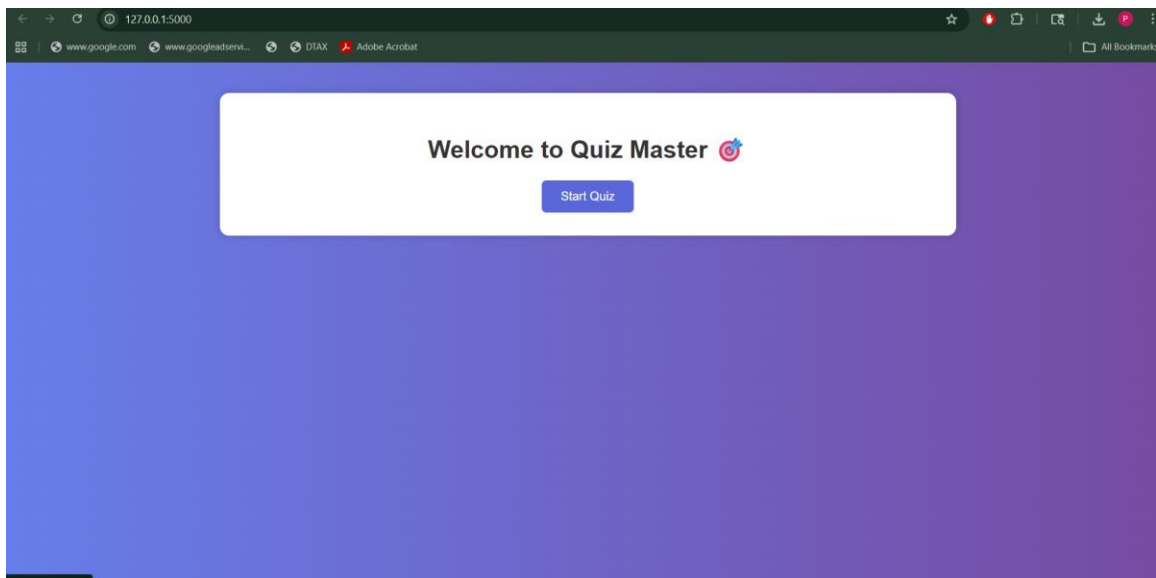
```
app.py | # style.css | index.html | result.html | ... | quiz.html
templates > index.html > html
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Quiz Master</title>
5     <link rel="stylesheet" href="/static/style.css">
6 </head>
7 <body>
8     <div class="container">
9         <h1>Welcome to Quiz Master </h1>
10        <a href="/quiz">
11            <button>Start Quiz</button>
12        </a>
13    </div>
14 </body>
15 </html>

templates > quiz.html > html
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Quiz</title>
5     <link rel="stylesheet" href="/static/style.css">
6 </head>
7 <body>
8
9     <div class="container">
10        <h2>Quiz Time </h2>
11
12        <form action="/result" method="POST">
13
14            {% for i in range(quiz|length) %}
15            <div class="question-box">
16                <p><b>{{ i+1 }}. {{ quiz[i].question }}</b></p>
17
18                {% for option in quiz[i].options %}
19                <label class="option">
20                    <!-- UNIQUE NAME FIX -->
21                    <input type="radio" name="q{{ i }}" />
22                    {{ option }}
23                </label>
24                {% endfor %}
25            </div>
26            {% endfor %}
27
28            <button type="submit">Submit Quiz</button>
29        </form>
30    </div>
31
32 </body>
```

```
# style.css X
static > # style.css > button:hover
1 body {
2   font-family: Arial;
3   background: linear-gradient(to right, #667e8b, #667e8b);
4   text-align: center;
5   color: #333;
6 }
7
8 .container {
9   background: white;
10  padding: 30px;
11  margin: 40px auto;
12  width: 60%;
13  border-radius: 12px;
14  box-shadow: 0 0 10px rgba(0,0,0,0.2);
15 }
16
17 .question-box {
18   margin: 20px 0;
19   text-align: left;
20 }
21
22 .option {
23   display: block;
24   margin: 8px 0;
25   padding: 8px;
26   background: #f1f1f1;
27   border-radius: 6px;
28   cursor: pointer;
29 }
30
31 .option:hover {
```

```
result.html X
templates > result.html > html > body > div.container > a
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Result</title>
5   <link rel="stylesheet" href="/static/style.css">
6 </head>
7 <body>
8   <div class="container">
9     <h2>Your Score 🎉</h2>
10
11     <h3>{{ score }} / {{ total }}</h3>
12
13     {% if score >= 3 %}
14       <p>Good Job 🍀</p>
15     {% else %}
16       <p>Try Again 😞</p>
17     {% endif %}
18
19     <a href="/quiz">
20       <button>Retake Quiz</button>
21     </a>
22   </div>
23 </body>
24 </html>
```

Output:



127.0.0.1:5000/quiz

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Quiz Time 🎉

1. CPU stands for?

- ☒ Central Processing Unit
- ☐ Control Unit
- ☐ Computer Unit
- ☐ None

2. Capital of India?

- ☒ Delhi
- ☐ Mumbai
- ☐ Chennai
- ☐ Kolkata

3. Flask is?

- ☒ Framework

127.0.0.1:5000/result

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Your Score 🎉

5 / 5

Good Job 🍀

[Retake Quiz](#)